- **Co-finite**: open $\iff$ **complements of finite sets**.
- **Co-countable**: open $\iff$ **complements of countable sets**.
- $\mathcal{B}$ gen. the coarsest containing itself.
- **Metric**: ≥ 0, **definite, symmetric, $\triangle$-inequality**.
- **Pseudo-metric**: $d(x,x) = 0$ but can be not definite.
- **Quasi-metric**: can be not symmetric.
- **Norm**: ≥ 0, **definite, $\triangle$-inequality**, $\|\lambda x\| = |\lambda| \|x\|$.
- **Distance between sets**: smallest pointwise distance.
- **Diameter of set**: sup of pointwise distance.
- L^p**-metric**: gen. the standard topo. on $\mathbb{R}^n$.

$$\max \|y_i - x_i\| \leq \left[\sum_{i=1}^{n} \|y_i - x_i\|^p\right]^{\frac{1}{p}} \leq n^{\frac{1}{p}} \max \|y_i - x_i\|.$$

- Metrics are equivalent $\iff c_1 d \leq d' \leq c_2 d$.
- $x \in \overline{A} \iff \forall$ open neighbourhood U of x, $U \cap A \neq \varnothing$.
- **Continuity**: $f^{-1}(S)$ is open $\forall S$ in sub-basis.
- TFAE:
  1. f is cont.;
  2. $\forall A \subseteq X$, $f(\overline{A}) \subseteq \overline{f(A)}$ (prove $\overline{A} \subseteq f^{-1}\left(\overline{f(A)}\right)$ cuz closed);
  3. for any closed set $B \subseteq Y$, $f^{-1}(B)$ is closed in X;
  4. $\forall x \in X$ and any open $V \subseteq Y$ with $f(x) \in V$, there is an open set $U \subseteq X$ s.t. $x \in U$ and $f(U) \subseteq V$.
- **Dense**: $\forall$ open U containing $x \in X$, $U \cap A \neq \varnothing$.
- Dense $\iff \overset{\circ}{X \setminus A} = \varnothing \iff \overline{A} = X$ (contra+).
- **Pasting lemma**: if $X = A \cup B$ for closed A, B and $f = g$ for all $x \in A \cap B$, then $h = x \in A ? f : (x \in B ? g)$ is cont. if $f \colon A \to Y$ and $g \colon B \to Y$ are.
- **Pull-back** topo.: $\left\{f^{-1}(U) : U \text{ is open}\right\}$ is the coarsest topo. ensuring cont. f. (quotient)
- **Uniform continuity**: $\forall \epsilon > 0$, there exists some $\delta > 0$ s.t. $d_X(x,y) < \delta \implies d_Y\left(f(x), f(y)\right) < \epsilon$.
- f is uni. cont. $\iff \forall \{x_i\}_i^\infty, \{y_i\}_i^\infty$ s.t. $\lim_{i\to\infty} d_X(x_i, y_i) = 0$, $\lim_{i\to\infty} d_Y\left(f(x_i), f(y_i)\right) = 0$.
- Limit of uni. convergent sequence is cont..
- $x \mapsto x^n$ converges to $\mathbf{1}_{x=1}$ pointwise but not uni..
- **Prod.** topo.: $\mathcal{S} \coloneqq \left\{\pi_{X_\alpha}^{-1}(U_\alpha) : \alpha \in \Lambda, U_\alpha \in \mathcal{T}_\alpha\right\}$.
- Prod. topo. is the coarsest for cont. π_α's.
- $x \mapsto (x)^{\mathbb{N}}$ is not cont. in the box topo..
- f is cont. iff all components are cont..
- Subspace topo. of prod. topo. equals prod. topo. of subspace topologies.
- Standard topo. on $\mathbb{R}^n$ is the prod. topo. by standard topologies on $\mathbb{R}^{m_i}$'s ($\implies$: project each open ball using radius $t = r - \|\boldsymbol{x} - \boldsymbol{y}\|$ and take finite prod. of projections; $\impliedby$: choose open ball with radius $r \coloneqq \min_{1\leq i\leq k} r_i$ and bound each component. Use BOX topo.).
- Prod. space basic open set: all but finitely many factors are not X_i.
- Box topo. is finer: e.g. $\prod_{i\in\mathbb{N}} U_i$.
- $d_1 = \sum_{i=1}^{n} d_{X_i}(x_i, y_i)$ and $d_\infty = \max d_{X_i}(x_i, y_i)$ both induce the prod. topo..
- $\rho(x,y) \coloneqq \frac{d(x,y)}{1+d(x,y)}$ is a metric with $\operatorname{diam}(X) < 1$.
- $D(\boldsymbol{x}, \boldsymbol{y}) \coloneqq \sup\left\{\frac{\rho_i(x_i, y_i)}{i} : i \in \mathbb{Z}^+\right\}$ is a metric inducing the infinite prod. topo..
- **Quotient map**: surj and U open iff $f^{-1}(U)$ open.
- Closed not open: $x \mapsto x \in [0,1] ? x : (x \in [2,3] ? x - 1)$.
- Open not closed: $x \mapsto x \in (0,1) ? x : (x \in (2,3) ? x - 1)$.
- Neither: $X \coloneqq \left([0,\infty) \times \mathbb{R}\right) \cup \left(\mathbb{R} \times \{0\}\right)$, $(x,y) \mapsto x$. **But this is a quotient map!**
- Both: id_X.
- **Open map**: cont. map from open set to open set.
- Surjective cont. + open or closed = quotient map.
- Quotient, open and closed maps are preserved under $\circ$.
- **Saturated**: $A = f^{-1}\left(f(A)\right)$ or $A = f^{-1}(S)$.
- Surjective cont. f is a quotient map iff $f(A)$ is open (closed) whenever A is saturated open (closed).
- Restriction to saturated sets preserves quotient map.
- Homeomorphism: bijective open map.
- Embedding: injective homeomorphism onto $f(X)$.
- Isometric: $d_Y(y_a, y_b) = d_X(a, b)$. Isometry if surj.

Countability

- 2nd countable $\implies \exists$ dense countable subset. $\impliedby$ is true if X is metrisable.
- $\left(\mathbb{R}, \mathcal{P}(\mathbb{R})\right)$ is 1st but not 2nd countable.
- Uncountable co-finite have no countable basis.
- $\exists$ nested countable basis $B_1 \subseteq B_2 \subseteq \cdots$.
- $x_n \to x \implies x \in \overline{X}$. $\impliedby$ is true if X is 1st countable.
- If f is cont., then for any sequence $f(x_i) \to f(x)$. The converse is true if X is first countable.
- Lindelof: every open cover has a countable sub-cover.
- 2nd countable $\implies$ Lindelof. $\impliedby$ if X is metrisable.

Separation

- Co-finite topo. is T_1 and T_2 if X is finite.
- $T_1 \iff \{x\}$ is closed $\forall x \in X$.
- **Embedding thm**: $\forall x$ and $\forall$ open U s.t. $x \in U$, $\exists \alpha \in \Lambda$ s.t. $f_\alpha(x) > 0$ and $f_\alpha(X \setminus U) = \{0\}$, then $F(x) = \left(f_\alpha(x)\right)_{\alpha\in\Lambda}$ is embedding.
- Metric spaces are T_2 so all finite subsets are closed.
- T_2 iff diagonal is closed.
- $T_3 \iff \forall x$ and $\forall$ open U s.t. $x \in U$, $\exists$ open V s.t. $x \in V$ and $\overline{V} \subseteq U$.
- $T_4 \iff \forall$ closed A and open U s.t. $A \subseteq U$, $\exists$ open V s.t. $A \subseteq V$ and $\overline{V} \subseteq U$.
- $T_4 \implies T_3 \implies T_2 \implies T_1$.
- Every metric space is T_4.
- Every 2nd countable T_3 space is T_4.
- Cont. f separates A, B: $f(A) = \{0\}$, $f(B) = \{1\}$.
- **Urysohn's lem**: Every T_4 space is completely normal.
- **Urysohn's metrisation thm**: every 2nd countable T_3 space is metrisable.

Compactness

- $\left[-\sqrt{2}, \sqrt{2}\right] \cap \mathbb{Q}$: closed and bounded, not cpt.
- Closed subset of cpt space is cpt.
- Cpt subset of T_2 space is closed.

- Set in co-finite space is cpt but closed iff it's finite.
- Closure of particular point is not cpt.
- Cpt $\implies (T_4 \iff T_3 \iff T_2)$
- Cont. f maps cpt set to cpt set.
- **Tube lemma**: If Y is cpt and $N \subseteq X \times Y$ open, contains $\{x_0\} \times Y$, then $\exists W \supseteq \{x_0\}$ open s.t. $W \times Y \subseteq N$.
- Y is cpt $T_2 \implies \Gamma_f$ closed $\iff f$ is cont..
- X cpt iff $\forall$ set of closed sets $\mathcal{G}$ with f.i.p, $\bigcap \mathcal{G} \neq \varnothing$.
- x is **isolated** $\iff \{x\}$ is open.
- If $U \neq \varnothing$ is open in a T_2 space and x is not isolated, then $\exists$ non-empty open $V \subseteq U$ s.t. $x \notin \overline{V}$.
- Non-empty T_2 space $\approx \mathbb{R}$ if has no isolated point.
- **Tychonoff's thm**: prod. of cpt is cpt.
- **Lim pt cpt**: every infinite $Y \subseteq X$ has a lim pt in X. Cpt $\implies$ lim pt cpt (set with no lim pt is finite).
- Let Y be trivial and $\mathbb{Z}^+$ be discrete, then $Y \times \mathbb{Z}^+$ is lim pt cpt but not cpt.
- **Seq cpt**: every sequence has a convergent subsequence. Seq cpt $\implies$ lim pt cpt.
- $\mathbb{R}$ basis (a, ∞)'s is lim pt ($\frac{a+a'}{2}$) but not seq ($\{-n\}$) cpt.
- **Leb num**: $\forall S$ with $\operatorname{diam}(S) < \delta$, $\exists U \in \mathcal{U}$ s.t. $S \subseteq U$.
- $(n, n+1)$-(± 1)-$\left(n \pm \frac{1}{n}\right)$ has no Leb num.
- Every open cover of seq cpt metric space has a Leb num (suppose not, $\frac{1}{n}$ are not Leb num, pick a seq).
- **Totally bounded**: $\forall \epsilon > 0$, $\exists$ finite cover by $B_\epsilon(x_i)$.
- Every seq cpt metric space is totally bounded (contra+ by recursive seq).
- Metric space: cpt iff lim pt cpt iff seq cpt (T_1, fix a seq with nested countable basis; Leb num & finite $\frac{\delta}{3}$-balls).
- $f\colon X \to Y$ is cont. in metric spaces. If X is cpt, then f is uni. cont. ($\frac{\epsilon}{2}$-ball cover Y, X seq cpt, Leb num).
- Totally bounded then $\operatorname{diam}(X) < \infty$ (max centre dist).
- $X \subseteq \mathbb{R}^n$ is totally bounded iff X is bounded ($\frac{\epsilon}{2}$-net).
- $\left(1 + \frac{1}{n}\right)^n$ converges to e.
- **Complete**: every Cauchy seq is convergent.
- $(\mathbb{R}^\omega, D)$ is complete (Cauchy's components are Cauchy).
- Subspace of complete metric space is complete iff closed (not closed $\implies$ not complete).
- Cpt iff complete and totally bounded (cover with 2^{-n}-balls, fix Cauchy subseq, prove seq cpt).
- **Heine-Borel Thm**: cpt in $\mathbb{R}^n$ iff closed and bounded.
- **Locally cpt**: $\exists$ cpt K and open U s.t. $x \in U \subseteq K$.
- X is locally cpt T_2 iff $\exists$ cpt T_2 Y and $h_Y\colon X \to Y$ homeomorphic onto $h_Y(X)$ and $Y = h_Y(X) \cup \{p\}$.
- Open sets in Y are open sets in X and complements of cpt sets in X. h_Y is just the identity.
- $\exists$ homeomorphism f s.t. $f \restriction_{h_Y(X)} = h_{Y'} \circ h_Y^{-1} \restriction_{h_Y(X)}$
- $\overline{\mathbb{D}}$ and $\mathbb{S}^2$ are the one-point cptification for $\mathbb{D}$ and $\mathbb{R}^n$ because $\boldsymbol{v} \mapsto \frac{\boldsymbol{v}}{1+\|\boldsymbol{v}\|}$ is bijection.
- A T_2 space is locally cpt iff $\forall x$ and open U s.t. $x \in U$, $\exists$ open cpt V s.t. $x \in V \subseteq U$.
- X locally cpt, then $A \subseteq X$ locally cpt if A closed or A open and X T_2.
- X locally cpt T_2 iff $\exists$ cpt T_2 Y s.t. X homeomorphic to open $A \subseteq Y$.
- **Uniform metric**: $\overline{\rho}(f, g) = \sup\left\{\rho\big(\pi_\alpha(f), \pi_\alpha(g)\big)\right\}$.
- Prod. $\subseteq$ Uniform $\subseteq$ Box
- Y is complete $\implies Y^X$ is complete.
- $\mathcal{C}(X, Y)$ and $\mathcal{B}(X, Y)$ are closed, and complete if Y is.
- **Supremum metric**: $d_{\sup}(f, g) = \sup d\big(f(x), g(x)\big)$.
- **Metric completion**: isometric ϕ s.t. $\overline{\phi(X)} = Y$.
- Every metric space has a unique metric completion up to isometry s.t. $f \restriction_{\phi(X)} = \phi' \circ \phi^{-1}$.
- **Topo. of cpt convergence**: gen'ed by $B(K, f, \epsilon) := \left\{g \in Y^X : d_{\sup}(f \restriction_K, g \restriction_K) < \epsilon\right\}$'s.
- **Cpt-open topo.**: gen'ed by $S(K, U) := \{g \in \mathcal{C}(X, Y) : g(K) \subseteq U\}$'s.
- Both are the same on $\mathcal{C}(X, Y)$.
- **Cptly gen'ed**: A open (closed) in X iff $A \cap K$ open (closed) in K for all cpt sets in X.
- Locally cpt or 1st countable $\implies$ cptly gen'ed.
- Cptly gen'ed $\implies$ (f cont. iff $f \restriction_K$ cont. $\forall$ cpt K).
- X cptly gen'ed $\implies \mathcal{C}(X, Y)$ closed under topo. of cpt convergence.
- **Equicont.**: $\exists$ open $U \forall \epsilon > 0$ s.t. $\forall f \in \mathcal{Y} \subseteq \mathcal{C}(X, Y)$, $d(y_{x_0}, y_x) < \epsilon \forall x \in U$.
- Totally bounded in $\overline{\rho} \implies$ equicont.
- Converge in prod. topo. = ptwise convergence.
- Equicont. family has the same topo. for cpt and ptwise convergence.
- **Arzela-Ascoli Thm**: $\mathcal{Y}_a := \{f(a) : f \in \mathcal{Y}\}$. $\overline{\mathcal{Y}} \subseteq \mathcal{C}(X, Y)$ is cpt if it's equicont. and $\overline{\mathcal{Y}_a}$ is cpt $\forall a \in X$. $\Longleftarrow$ is true if X is locally cpt T_2.

Connectedness

- Connected iff only X and $\varnothing$ are clopen (contra+).
- $X = (U, V)$, $Y \subseteq X$ connected $\implies Y \subseteq U$ or $Y \subseteq V$ (contra+).
- Connected sets with $\cap \neq \varnothing$ has connected $\cup$ (suppose not and condition on $A_\alpha \setminus U$).
- A connected and $A \subseteq B \subseteq \overline{A} \implies B$ connected (suppose not).
- Cont. f maps connected to connected (contra+).
- Prod. of connected is connected (contra+).
- Path connected $\implies$ connected (contra+, suppose exists cont. f).
- Path component $\subseteq$ cc.
- **Locally connected**: $\forall$ open U s.t. $x \in U$, $\exists$ open connected V s.t. $x \in V \subseteq U$.
- S: graph of $\sin \frac{2\pi}{x}$ at $(0, 1]$. $\overline{S} = S \cup \{0\} \times [-1, 1]$ is connected, but not path/locally connected (no $(0,0)$-(x, y) path: $\max f^{-1}(\{0\} \times [-1, 1]) \mapsto y_0 := \lim y(t)$; not locally connected: 0.5-ball at $\mathbf{0}$ contains a δ-square, $\exists x \in (0, \delta)$ s.t. $\sin = 1$, split there).
- Locally connected iff every cc of every open set is open.
- CCs and path components are the same in locally path connected space (suppose not but a path cc $\subseteq$ a cc).
- Quotient topo. on $\tilde{X}$: cc's of locally path connected space, is discrete (prove $\mathcal{P}\left(\tilde{X}\right)$ is a quotient topo.).
- Locally path connected $\implies$ locally connected (counter e.g. empty graph).